Multimodal AI (MAS.S60, 6.S985)

Project Final

The final assignment is separated into two major parts: a final presentation and a final report. **Please be sure to read the instructions for the final report BEFORE reading the final instructions.**

Final Presentations - **Class time on Tuesday May 12th, 2026, at E14 3rd or 6th floor atrium (TBD)**

Final Report - **Deadline: Tuesday May 19th, 2026 at 11:59pm ET**

Final Report Instructions

The final project assignment has two parts: presentation and report. This document details the instructions for the report. The final report should summarize the main results of your course project. The final report should follow the conventional layout for research papers with an abstract, introduction, related work, proposed model, experiments, results, discussion, and conclusion. While multiple research ideas may have been explored during the semester, the final report can focus on only a subset of them. In other words, instead of describing many research ideas, the final report can focus on 1 or 2 research ideas that worked best and present them in detail. The main results may be positive or negative (e.g., the model did not end up improving the previous approaches), but the most important part is to discuss these results and follow them with good error analysis to understand why it succeeded (or not). The report should end with a discussion of potential future work.

Formatting and page limits

Please format your proposal report using the NeurIPS format (link). The maximum number of pages is 9 pages + unlimited extra pages for references. You can optionally include an appendix document with extra details about your project.

Github repository

Github repository (20 points)

It is required that you create and maintain your own GitHub repository for this course, which will serve as the homebase for your work: a living, evolving portfolio that showcases your weekly ideas, experiments, and technical growth over time. This repository should bring everything together in one place: past assignments, ongoing explorations, and final projects. Each repository must be clearly organized, publicly readable within the class, and include a top-level README.md, a README.md for each assignment, and all completed work so far. For the final project, each team member should include their own copy of the project in their repository. Feel free to use a modified copy of this template. Think of this repository not just as a requirement, but as something you would be proud to share with classmates and people taking this class in the future who can get inspiration and help from. You are encouraged to “bedazzle” it: add visuals, diagrams, results, notes, side experiments, and anything else that makes your work more engaging, understandable and personal to you. A well-crafted repository tells the story of how you think, what you’ve explored, and how your ideas evolve. Over the semester, this space will grow into a rich technical lab notebook and creative portfolio: one that reflects both your curiosity and your ability to build with multimodal AI.

Equal contributions between teammates

It is important that all team members participate equally in the project. The Github repository may be used to help assess the relative work performed by each member of the team. If any team member has concerns about the participation level of other members, they should contact the instructor and/or TA as promptly as possible.

Assignment report outline

- **Abstract (1 long paragraph, 5 points):** In a few sentences, you should motivate your problem, explain the specific technical challenge you are solving, describe the main contributions of your approach, and summarize the main findings.
- **Introduction (about 3/4 page to 1 page, 10 points):** First, describe and motivate your research problem. Explain why this problem is important for the research community and, if possible, society in general. Second, define the specific research challenge(s) that have not been addressed in prior work. Third, explain your proposed idea(s) and how they will help solve (or better understand) the specific research challenges. This third paragraph often starts with "In this paper,…". If possible, have an overview figure illustrating the specific research challenges and how you propose to solve them.
- **Related work (about a page, 5 points):** The goal of this section is twofold: (1) present an overview of the work happening in your research area, and (2) highlight how your approach differs from prior work. Instead of simply listing all the prior work, try to group it by category or research problem. This will allow you to have a more structured related work section and will make it easier for the reader to place your work in the right context. This section should include about 15-20 citations of prior work. Usually, in the last paragraph of this section, you will have a paragraph stating how your work differs from the prior work (e.g., "To our knowledge, this paper is the first to…").
- **Problem statement (1/4 to 1/2 of a page, 5 points):** Mathematically formalize your research problem. This should include the mathematical definition of the variables involved in your problem (e.g., input variable x and labels y). You should also include an objective function describing your main objective. While this section may be relatively short (e.g., a quarter of a page), you should be consistent with your notation later in the paper.
- **Proposed method (about 2 pages, 20 points):** Describe in detail the main contributions of your proposed method. Most final reports will focus on only one proposed model, but larger teams can include 2 proposed approaches (research ideas). Include at least one figure to illustrate the main novelty of the proposal approach. Try to formalize the idea and proposed model as well as possible, with mathematical formalism as often as possible (e.g., define the loss function of your proposed model). Follow the same notation as the problem statement, whenever possible. If possible, describe how the model is optimized and how inference is performed.
- **Experimental methodology (about 1/2 to 1 page, 10 points):** Describe the dataset you are using for this project. Describe the input modalities and annotations available in this dataset. Describe your experimental methodology for evaluating the multimodal baseline model(s). This should include the way you split your dataset for training, validation, and testing. It should also include any details about the hyperparameters and the evaluation metrics. If the space is too short, you can move some of the implementation details to the appendix, but the main design decision

should be explained in the main paper. Grading will only be done using the main part of the paper. Appendix supplementary material is optional for grading.

- **Results and discussion (about 3 pages, 35 points):** This is an important part of the report. As often as possible, present your experimental results in tables and figures. This section should include more than re-running experiments from pre-existing baseline models. The goal of the course project is to explore new research directions. You should include a detailed discussion of your results, including an analysis of failure cases whenever possible. Discussion and error analysis are important parts of the report. You should have an intuition on why your approach succeeded (or failed). Negative results are still very interesting results as long as we have an intuition on why the model did not perform well.
- **Conclusion and future directions (about 1 page, 10 points):** State the main contributions of the paper. If possible, discuss some of the limitations of the current approach. Include a discussion of future research directions.

Final Presentation Instructions

The proposal will also include a poster presentation that will be open to the MIT community. The presentations should be self-contained so that attendees can understand the research problem and the main novelty of your course project.

- While we expect everyone to mingle around and look at each other's posters, the instructors and TAs will stop by each poster and grade each team's poster presentation.
- The time limit for each poster presentation is 2 minutes per student in the group (i.e., a 3-person group would present for 6 minutes). **All team members are expected to participate in both the creation of the poster and the presentation itself.**
- Grading will be based on 5 criteria **(20 points each for 100 points total)**:
    - Good motivation for the problem and existing literature
    - Good motivation for the new proposed ideas and how they address gaps in existing work
    - New method and experiments with clear explanation of main results and analysis
    - Presentation skills - clear explanations and nonverbal presentation skills
    - Quality of the poster with good visuals and not too much text

Suggestions for the presentation:

- Motivate and illustrate your research problem. If possible, include a visual example (e.g., image or short video). Emphasize the technical challenge that your model will address in this research problem.
- Present your main contribution. While you may not be able to cover all the details of your idea, be sure to give a high-level intuition about the novelty of your approach.
- Present 1 or 2 key results from your experiments. Do not try to present all your results. Explain your results (either positive or negative) with some extra analysis. Why did it succeed (or fail)?
- If you have time, you can discuss future research directions.